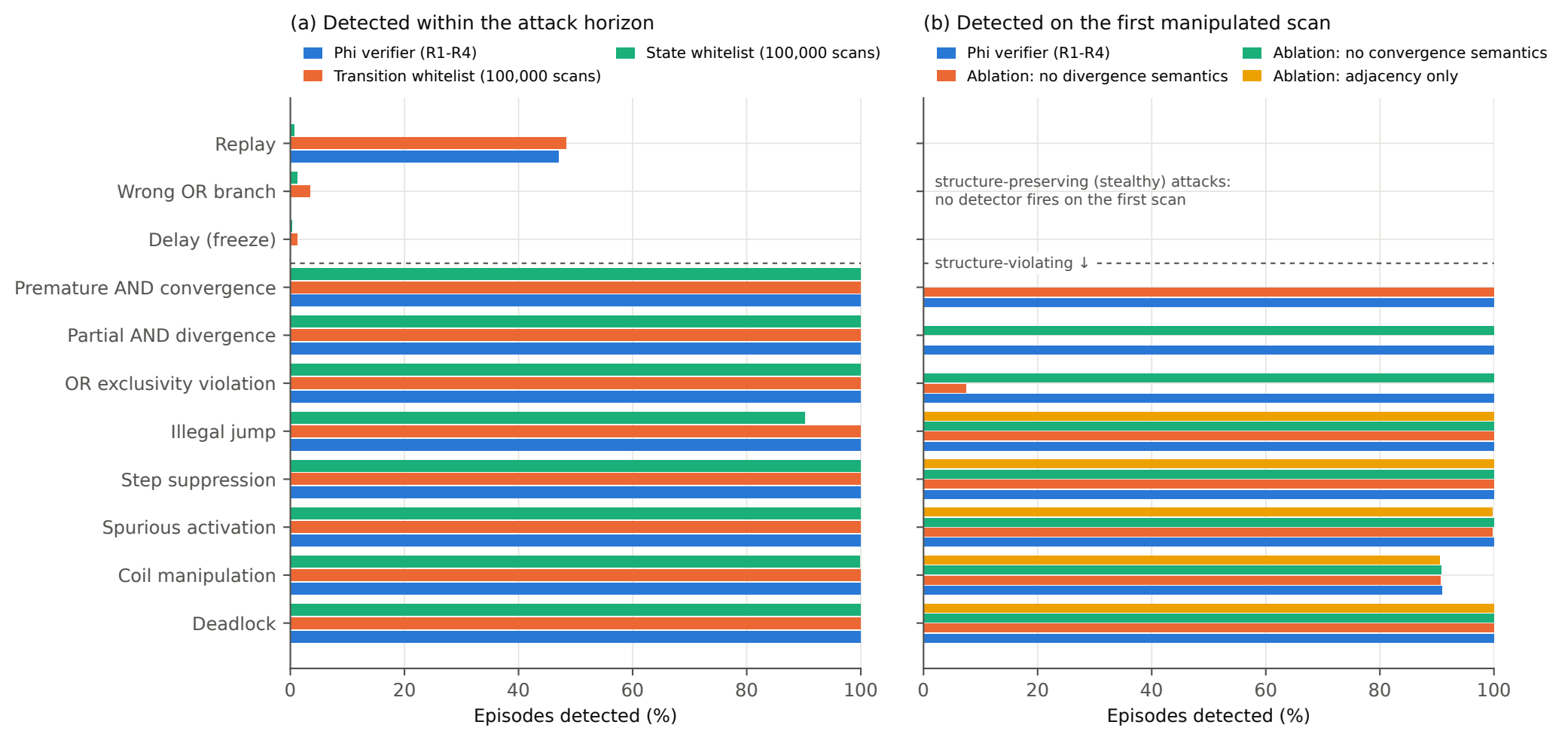
(a) Detected within the attack horizon
Phi verifier (R1-R4)
Transition whitelist (100,000 scans)
State whitelist (100,000 scans)
Replay
Wrong OR branch
Delay (freeze)
Premature AND convergence
Partial AND divergence
OR exclusivity violation
Illegal jump
Step suppression
Spurious activation
Coil manipulation
Deadlock
0
20
40
60
80
100
Episodes detected (%)
(b) Detected on the first manipulated scan
Phi verifier (R1-R4)
Ablation: no divergence semantics
Ablation: no convergence semantics
Ablation: adjacency only
structure-preserving (stealthy) attacks:
no detector fires on the first scan
structure-violating ↓
0
20
40
60
80
100
Episodes detected (%)